

A Markov Chain Problem

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This problem below comes from one of the questions of the first MATH 632 midterm exam taught by Professor Benedek Valko in Fall 2025. I am in debt to Prof. Valko in many ways, and I remark solving this problem on that midterm as one of the most amazing moments during my undergrad. Moreover, the course, MATH 632, is the only course I ever listened from start to end though skipping one, it changed my old point of view lasting for many years that one could only learn math through self study.

The Problem

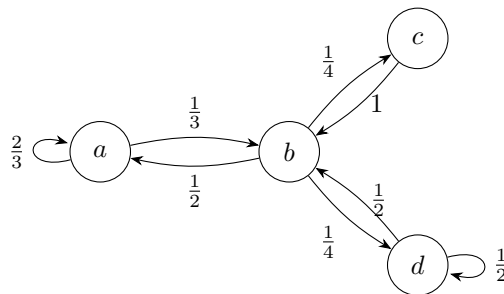
Consider the Markov chain on the state space

$$S = \{a, b, c, d\}$$

given by the transition diagram below. Find

$$\mathbb{P}_b(T_a < T_c),$$

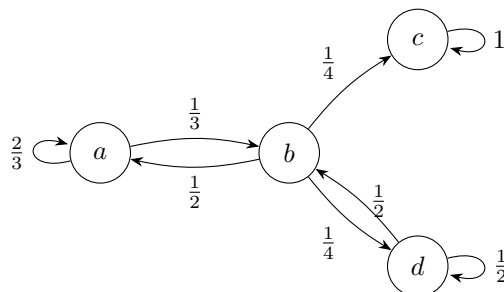
the probability that the Markov chain started from b visits a before c .



My Solution During the Exam.

It amounts to making c absorbing.

Hence the transition diagram becomes



We have

$$\begin{aligned}
\mathbb{P}_b(T_a < T_c) &= \mathbb{P}_b(T_a < T_c \mid X_1 = c) P(b, c) \\
&\quad + \mathbb{P}_b(T_a < T_c \mid X_1 = d) P(b, d) \\
&\quad + \mathbb{P}_b(T_a < T_c \mid X_1 = a) P(b, a) \\
&= 0 + P(b, a) + \frac{1}{4} \mathbb{P}_b(T_a < T_c \mid X_1 = d) \\
&= \frac{1}{2} + \frac{1}{4} \mathbb{P}_d(T_a < T_c).
\end{aligned}$$

Now,

$$\begin{aligned}
\mathbb{P}_d(T_a < T_c) &= \mathbb{P}_d(T_a < T_c \mid X_1 = d) P(d, d) \\
&\quad + \mathbb{P}_d(T_a < T_c \mid X_1 = b) P(d, b) \\
&= \frac{1}{2} \mathbb{P}_d(T_a < T_c) + \frac{1}{2} \mathbb{P}_b(T_a < T_c).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathbb{P}_b(T_a < T_c) &= \frac{1}{2} + \frac{1}{4} \left(\frac{1}{2} \mathbb{P}_d(T_a < T_c) + \frac{1}{2} \mathbb{P}_b(T_a < T_c) \right) \\
&= \frac{1}{2} + \frac{1}{4} \left[\frac{1}{2} \left(\frac{1}{2} \mathbb{P}_d(T_a < T_c) + \frac{1}{2} \mathbb{P}_b(T_a < T_c) \right) + \frac{1}{2} \mathbb{P}_b(T_a < T_c) \right].
\end{aligned}$$

Continuing this expansion,

$$\mathbb{P}_d(T_a < T_c) = \sum_{h=1}^n \frac{1}{2^h} \mathbb{P}_b(T_a < T_c) + \frac{1}{2^n} \mathbb{P}_d(T_a < T_c).$$

Let $n \rightarrow \infty$. Since

$$\frac{1}{2^n} \mathbb{P}_d(T_a < T_c) \rightarrow 0$$

and

$$\sum_{h=1}^{\infty} \frac{1}{2^h} = 1,$$

we obtain

$$\mathbb{P}_d(T_a < T_c) = \mathbb{P}_b(T_a < T_c).$$

Hence

$$\mathbb{P}_b(T_a < T_c) = \frac{1}{2} + \frac{1}{4} \mathbb{P}_b(T_a < T_c),$$

so

$$\frac{3}{4} \mathbb{P}_b(T_a < T_c) = \frac{1}{2}.$$

Therefore,

$$\boxed{\mathbb{P}_b(T_a < T_c) = \frac{2}{3}}.$$

Why The Problem Moved Me

What I found beautiful about this problem only became clear to me after the exam. During the exam, I did not remember the standard matrix method for absorbing Markov chains. I simply followed the probabilistic structure of the chain: starting from state d , the process either returns to b with probability $1/2$, or stays at d with probability $1/2$. This gives the recursion

$$p_d = \frac{1}{2}p_b + \frac{1}{2}p_d,$$

where $p_x = \mathbb{P}_x(T_a < T_c)$. Instead of immediately solving this linear equation algebraically, I repeatedly substituted the expression for p_d into itself:

$$p_d = \frac{1}{2}p_b + \frac{1}{2}\left(\frac{1}{2}p_b + \frac{1}{2}p_d\right) = \left(\frac{1}{2} + \frac{1}{2^2}\right)p_b + \frac{1}{2^2}p_d,$$

and, after n repetitions,

$$p_d = \left(\sum_{k=1}^n \frac{1}{2^k}\right)p_b + \frac{1}{2^n}p_d.$$

Since $0 \leq p_d \leq 1$, the remaining term satisfies

$$\frac{1}{2^n}p_d \longrightarrow 0,$$

while

$$\sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$$

Thus I strikingly found that

$$p_d = p_b.$$

After the exam, a classmate told me that the intended solution was the matrix formula

$$U = (I - Q)^{-1}S.$$

Only then did I realize what I had accidentally reconstructed. For a transient matrix Q ,

$$(I - Q)^{-1} = I + Q + Q^2 + Q^3 + \cdots,$$

which is precisely the **Neumann series**!!! The factor $2^{-n}p_d$ that I watched vanish was, in miniature, the same phenomenon as the transient contribution Q^n disappearing as $n \rightarrow \infty$. What moved me was that I had not consciously recalled the theorem at all. I had simply followed the structure of the problem far enough that the theorem re-emerged on its own!